Component-I: LSTM Sequence Generation

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# ================================
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import numpy as np
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense

# ------------------------------
# Task 1: Load Dataset
# ------------------------------
data = """machine learning models learn patterns from data.
sequence models process data step by step.
recurrent neural networks are designed for sequential tasks.
rnn models maintain hidden states across time steps.

long short term memory networks solve long dependency problems.
lstm uses gates to control information flow.
gru models simplify the lstm architecture.
sequence prediction is useful in many applications.

language modeling predicts the next word in a sentence.
speech recognition processes audio sequences.
time series forecasting predicts future values.
music generation creates new melodies.

generative models learn probability distributions.
they generate new samples similar to training data.
sequence generation is widely used in artificial intelligence.
deep learning improves sequence modeling performance.
"""

# ------------------------------
# Task 2: Tokenization
# ------------------------------
tokenizer = Tokenizer()
tokenizer.fit_on_texts([data])
total_words = len(tokenizer.word_index) + 1

# ------------------------------
# Task 3: Create Sequences
# ------------------------------
input_sequences = []

for line in data.split('\n'):
    token_list = tokenizer.texts_to_sequences([line])[0]
```

```
    for i in range(1, len(token_list)):
        input_sequences.append(token_list[:i+1])

# Padding
max_len = max(len(seq) for seq in input_sequences)
input_sequences = np.array(
    pad_sequences(input_sequences, maxlen=max_len, padding='pre')
)

# Split into X and y
X = input_sequences[:, :-1]
y = input_sequences[:, -1]

# ------------------------------
# Task 4: Build LSTM Model
# ------------------------------
model = Sequential([
    Embedding(total_words, 100, input_length=max_len-1),
    LSTM(150),
    Dense(total_words, activation='softmax')
])

model.compile(loss='sparse_categorical_crossentropy',
optimizer='adam')

# ------------------------------
# Task 5: Train Model
# ------------------------------
model.fit(X, y, epochs=200, verbose=1)

# ------------------------------
# Task 6: Generate Text
# ------------------------------
def generate_text(seed_text, next_words):
    for _ in range(next_words):
        token_list = tokenizer.texts_to_sequences([seed_text])[0]
        token_list = pad_sequences(
            [token_list], maxlen=max_len-1, padding='pre'
        )

        predicted = np.argmax(model.predict(token_list), axis=-1)

        for word, index in tokenizer.word_index.items():
            if index == predicted:
                seed_text += " " + word
                break

    return seed_text

# Example Output
```

```
print("\nGenerated Text (LSTM):")
print(generate_text("machine learning", 6))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
embedding.py:100: UserWarning: Argument `input_length` is deprecated.
Just remove it.
  warnings.warn(

Epoch 1/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 4s 22ms/step - loss: 4.4649
Epoch 2/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 4.4478
Epoch 3/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 4.4278
Epoch 4/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 4.3976
Epoch 5/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 4.3403
Epoch 6/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 4.2569
Epoch 7/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 4.2336
Epoch 8/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 4.2054
Epoch 9/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 4.1581
Epoch 10/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 4.1393
Epoch 11/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step - loss: 4.1052
Epoch 12/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 4.0733
Epoch 13/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 4.0371
Epoch 14/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 3.9856
Epoch 15/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 3.9457
Epoch 16/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 3.8852
Epoch 17/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 3.8243
Epoch 18/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 3.7541
Epoch 19/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 3.6780
Epoch 20/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 3.6155
Epoch 21/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 3.5494
```

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Epoch 22/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 3.4843
Epoch 23/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 3.3670
Epoch 24/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 3.3111
Epoch 25/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 3.1964
Epoch 26/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 3.1214
Epoch 27/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 3.0237
Epoch 28/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.9225
Epoch 29/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.8101
Epoch 30/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.7277
Epoch 31/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.6233
Epoch 32/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.5109
Epoch 33/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.3983
Epoch 34/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.3006
Epoch 35/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.2027
Epoch 36/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.1158
Epoch 37/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 2.0186
Epoch 38/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 1.9349
Epoch 39/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 1.8627
Epoch 40/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 1.7772
Epoch 41/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 1.6950
Epoch 42/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 1.6394
Epoch 43/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 1.5641
Epoch 44/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 1.5031
Epoch 45/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 1.4517
Epoch 46/200
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3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 1.3862
Epoch 47/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - loss: 1.3421
Epoch 48/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 1.2828
Epoch 49/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 46ms/step - loss: 1.2422
Epoch 50/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - loss: 1.1912
Epoch 51/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 1.1454
Epoch 52/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 1.1111
Epoch 53/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.0726
Epoch 54/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.0393
Epoch 55/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.0046
Epoch 56/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.9742
Epoch 57/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.9350
Epoch 58/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.9060
Epoch 59/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.8771
Epoch 60/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.8421
Epoch 61/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.8236
Epoch 62/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.7919
Epoch 63/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.7697
Epoch 64/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.7484
Epoch 65/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.7254
Epoch 66/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.6996
Epoch 67/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.6841
Epoch 68/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.6628
Epoch 69/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.6424
Epoch 70/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.6283
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Epoch 71/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.6054
Epoch 72/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.5898
Epoch 73/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.5715
Epoch 74/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.5595
Epoch 75/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.5412
Epoch 76/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.5263
Epoch 77/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.5112
Epoch 78/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.4972
Epoch 79/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.4842
Epoch 80/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.4696
Epoch 81/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.4583
Epoch 82/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4454
Epoch 83/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.4329
Epoch 84/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.4203
Epoch 85/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.4130
Epoch 86/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.4049
Epoch 87/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.3910
Epoch 88/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.3803
Epoch 89/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.3718
Epoch 90/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.3641
Epoch 91/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.3500
Epoch 92/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.3451
Epoch 93/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.3345
Epoch 94/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.3286
Epoch 95/200
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3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.3179
Epoch 96/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.3115
Epoch 97/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.3049
Epoch 98/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.2949
Epoch 99/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2903
Epoch 100/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2812
Epoch 101/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2739
Epoch 102/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2683
Epoch 103/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2623
Epoch 104/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2541
Epoch 105/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2501
Epoch 106/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.2446
Epoch 107/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2397
Epoch 108/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2320
Epoch 109/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2284
Epoch 110/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2247
Epoch 111/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2171
Epoch 112/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2125
Epoch 113/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2083
Epoch 114/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.2045
Epoch 115/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.2001
Epoch 116/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1953
Epoch 117/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1912
Epoch 118/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1876
Epoch 119/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1842
```

```
Epoch 120/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1796
Epoch 121/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1764
Epoch 122/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1737
Epoch 123/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1696
Epoch 124/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1664
Epoch 125/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1642
Epoch 126/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1620
Epoch 127/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1591
Epoch 128/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1554
Epoch 129/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1525
Epoch 130/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1497
Epoch 131/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1480
Epoch 132/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.1454
Epoch 133/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1431
Epoch 134/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1408
Epoch 135/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1394
Epoch 136/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1371
Epoch 137/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1344
Epoch 138/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1328
Epoch 139/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1304
Epoch 140/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1285
Epoch 141/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1267
Epoch 142/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1248
Epoch 143/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1227
Epoch 144/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1219
Epoch 145/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1205
Epoch 146/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1195
Epoch 147/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1162
Epoch 148/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1154
Epoch 149/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1140
Epoch 150/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1128
Epoch 151/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1107
Epoch 152/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1097
Epoch 153/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.1082
Epoch 154/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1070
Epoch 155/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.1063
Epoch 156/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1051
Epoch 157/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1034
Epoch 158/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1029
Epoch 159/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.1013
Epoch 160/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1003
Epoch 161/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0995
Epoch 162/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0978
Epoch 163/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0969
Epoch 164/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0955
Epoch 165/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0951
Epoch 166/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0945
Epoch 167/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0932
Epoch 168/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0926
```

```
Epoch 169/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0913
Epoch 170/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0906
Epoch 171/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0898
Epoch 172/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0891
Epoch 173/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0880
Epoch 174/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.0873
Epoch 175/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0869
Epoch 176/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0863
Epoch 177/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0857
Epoch 178/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0846
Epoch 179/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0840
Epoch 180/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0827
Epoch 181/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0825
Epoch 182/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0814
Epoch 183/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0814
Epoch 184/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0813
Epoch 185/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0808
Epoch 186/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0798
Epoch 187/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0792
Epoch 188/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0789
Epoch 189/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0780
Epoch 190/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0777
Epoch 191/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0767
Epoch 192/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0769
Epoch 193/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0760
Epoch 194/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0758
Epoch 195/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0745
Epoch 196/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0748
Epoch 197/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0732
Epoch 198/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0740
Epoch 199/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.0735
Epoch 200/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0726

Generated Text (LSTM):
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 186ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
machine learning models learn patterns from data data
```

```python
# Example Output
print("\nGenerated Text (LSTM):")
print(generate_text("machine learning", 6))
```

```
Generated Text (LSTM):
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
machine learning models learn patterns from data step
```

```python
# Example Output
print("\nGenerated Text (LSTM):")
print(generate_text("long short term", 6))
```

```
Generated Text (LSTM):
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
long short term memory networks solve long dependency problems
```

Component-II: Transformer Sequence Generation

```
# ================================
# Component-II: Transformer Sequence Generation
# ================================

import tensorflow as tf
from tensorflow.keras.layers import Embedding, Dense,
LayerNormalization, Dropout
from tensorflow.keras.models import Model
import numpy as np

# -------------------------------
# Task 1 & 2: Dataset + Tokenization
# -------------------------------
tokenizer = Tokenizer()
tokenizer.fit_on_texts([data])
total_words = len(tokenizer.word_index) + 1

input_sequences = []
for line in data.split('\n'):
    token_list = tokenizer.texts_to_sequences([line])[0]
    for i in range(1, len(token_list)):
        input_sequences.append(token_list[:i+1])

max_len = max(len(seq) for seq in input_sequences)
input_sequences = pad_sequences(input_sequences, maxlen=max_len,
padding='pre')

X = input_sequences[:, :-1]
y = input_sequences[:, -1]

# -------------------------------
# Task 3: Positional Encoding
# -------------------------------
class PositionalEncoding(tf.keras.layers.Layer):
    def __init__(self, max_len, d_model):
        super().__init__()
        self.pos_encoding = self.positional_encoding(max_len, d_model)

    def positional_encoding(self, position, d_model):
        angles = np.arange(position)[:, np.newaxis] / np.power(
            10000, (2 * (np.arange(d_model)[np.newaxis, :] // 2)) /
d_model
        )
        pos_encoding = angles.copy()
        pos_encoding[:, 0::2] = np.sin(angles[:, 0::2])
```

```
        pos_encoding[:, 1::2] = np.cos(angles[:, 1::2])
        return tf.cast(pos_encoding[np.newaxis, ...],
dtype=tf.float32)

    def call(self, x):
        return x + self.pos_encoding[:, :tf.shape(x)[1], :]

# -------------------------------
# Task 4: Transformer Block
# -------------------------------
class TransformerBlock(tf.keras.layers.Layer):
    def __init__(self, embed_dim, num_heads, ff_dim):
        super().__init__()
        self.att =
tf.keras.layers.MultiHeadAttention(num_heads=num_heads,
key_dim=embed_dim)
        self.ffn = tf.keras.Sequential([
            Dense(ff_dim, activation='relu'),
            Dense(embed_dim),
        ])
        self.norm1 = LayerNormalization()
        self.norm2 = LayerNormalization()
        self.dropout1 = Dropout(0.1)
        self.dropout2 = Dropout(0.1)

    def call(self, inputs):
        attn_output = self.att(inputs, inputs)
        out1 = self.norm1(inputs + self.dropout1(attn_output))
        ffn_output = self.ffn(out1)
        return self.norm2(out1 + self.dropout2(ffn_output))

# -------------------------------
# Build Transformer Model
# -------------------------------
embed_dim = 64
num_heads = 2
ff_dim = 128

inputs = tf.keras.Input(shape=(max_len-1,))
x = Embedding(total_words, embed_dim)(inputs)
x = PositionalEncoding(max_len, embed_dim)(x)
x = TransformerBlock(embed_dim, num_heads, ff_dim)(x)
x = tf.keras.layers.GlobalAveragePooling1D()(x)
outputs = Dense(total_words, activation='softmax')(x)

model = Model(inputs, outputs)

model.compile(loss='sparse_categorical_crossentropy',
optimizer='adam')
```

```python
# ------------------------------
# Task 5: Train
# ------------------------------
model.fit(X, y, epochs=200, verbose=1)

# ------------------------------
# Task 6: Generate Text
# ------------------------------
def generate_text_transformer(seed_text, next_words):
    for _ in range(next_words):
        token_list = tokenizer.texts_to_sequences([seed_text])[0]
        token_list = pad_sequences(
            [token_list], maxlen=max_len-1, padding='pre'
        )

        predicted = np.argmax(model.predict(token_list), axis=-1)

        for word, index in tokenizer.word_index.items():
            if index == predicted:
                seed_text += " " + word
                break

    return seed_text

print("\nGenerated Text (Transformer):")
print(generate_text_transformer("deep learning", 6))
```

```
Epoch 1/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 10s 1s/step - loss: 4.7923
Epoch 2/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - loss: 4.5582
Epoch 3/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.4654
Epoch 4/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 4.4076
Epoch 5/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 4.3909
Epoch 6/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 4.3504
Epoch 7/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.3271
Epoch 8/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.2974
Epoch 9/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.2701
Epoch 10/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 4.2583
Epoch 11/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 4.2696
Epoch 12/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.2507
Epoch 13/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.2422
Epoch 14/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.2116
Epoch 15/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.2142
Epoch 16/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 4.2095
Epoch 17/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.1967
Epoch 18/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 4.1682
Epoch 19/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.1548
Epoch 20/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 4.1387
Epoch 21/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 4.1176
Epoch 22/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 4.0932
Epoch 23/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 4.0672
Epoch 24/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 4.0266
Epoch 25/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.9868
Epoch 26/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 3.9372
Epoch 27/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.8986
Epoch 28/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.8311
Epoch 29/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.7657
Epoch 30/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 3.6908
Epoch 31/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.6268
Epoch 32/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.5572
Epoch 33/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.4470
Epoch 34/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.3602
Epoch 35/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.3087
Epoch 36/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 3.1869
```

```
Epoch 37/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 3.1248
Epoch 38/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.9938
Epoch 39/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.9344
Epoch 40/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.8469
Epoch 41/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.7274
Epoch 42/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.6061
Epoch 43/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.5327
Epoch 44/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.4455
Epoch 45/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.3636
Epoch 46/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.2225
Epoch 47/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 2.1622
Epoch 48/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 2.0420
Epoch 49/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.9816
Epoch 50/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.8884
Epoch 51/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.8035
Epoch 52/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.7145
Epoch 53/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 1.6031
Epoch 54/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 1.5777
Epoch 55/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 1.4748
Epoch 56/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.4118
Epoch 57/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 1.3515
Epoch 58/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.2971
Epoch 59/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.2493
Epoch 60/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.2014
Epoch 61/200
```

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3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.1080
Epoch 62/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.0804
Epoch 63/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 1.0380
Epoch 64/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.9721
Epoch 65/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.9152
Epoch 66/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.8993
Epoch 67/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.8453
Epoch 68/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.8325
Epoch 69/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.7980
Epoch 70/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.7513
Epoch 71/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.7139
Epoch 72/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.7021
Epoch 73/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.6650
Epoch 74/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.6423
Epoch 75/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.6005
Epoch 76/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.5802
Epoch 77/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.5511
Epoch 78/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.5279
Epoch 79/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.5295
Epoch 80/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.4873
Epoch 81/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.4699
Epoch 82/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.4646
Epoch 83/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.4316
Epoch 84/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.4194
Epoch 85/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.3924
```

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Epoch 86/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.3817
Epoch 87/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.3722
Epoch 88/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.3658
Epoch 89/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.3364
Epoch 90/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.3300
Epoch 91/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.3189
Epoch 92/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.3170
Epoch 93/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.3018
Epoch 94/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2955
Epoch 95/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2761
Epoch 96/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2655
Epoch 97/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2626
Epoch 98/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.2488
Epoch 99/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.2544
Epoch 100/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2324
Epoch 101/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2424
Epoch 102/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.2268
Epoch 103/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2263
Epoch 104/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2145
Epoch 105/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2101
Epoch 106/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.2154
Epoch 107/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.2033
Epoch 108/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.2050
Epoch 109/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.2044
Epoch 110/200
```

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3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1925
Epoch 111/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1952
Epoch 112/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1814
Epoch 113/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1787
Epoch 114/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1680
Epoch 115/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1617
Epoch 116/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1605
Epoch 117/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 0.1563
Epoch 118/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1510
Epoch 119/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1460
Epoch 120/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.1456
Epoch 121/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1496
Epoch 122/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1376
Epoch 123/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1391
Epoch 124/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.1380
Epoch 125/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1263
Epoch 126/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1312
Epoch 127/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1241
Epoch 128/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1252
Epoch 129/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1232
Epoch 130/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1236
Epoch 131/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1184
Epoch 132/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1172
Epoch 133/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1196
Epoch 134/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1092
```

```
Epoch 135/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1132
Epoch 136/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1067
Epoch 137/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1049
Epoch 138/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1114
Epoch 139/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1014
Epoch 140/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1078
Epoch 141/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1080
Epoch 142/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1014
Epoch 143/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1020
Epoch 144/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0927
Epoch 145/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1016
Epoch 146/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0973
Epoch 147/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0865
Epoch 148/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0916
Epoch 149/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.1101
Epoch 150/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0965
Epoch 151/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0835
Epoch 152/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0911
Epoch 153/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.1010
Epoch 154/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0857
Epoch 155/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0889
Epoch 156/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0870
Epoch 157/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0843
Epoch 158/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0829
Epoch 159/200
```

```
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0864
Epoch 160/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0864
Epoch 161/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0887
Epoch 162/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0827
Epoch 163/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0826
Epoch 164/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0810
Epoch 165/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0720
Epoch 166/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0773
Epoch 167/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0763
Epoch 168/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0742
Epoch 169/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0728
Epoch 170/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0814
Epoch 171/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0728
Epoch 172/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0733
Epoch 173/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0804
Epoch 174/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0763
Epoch 175/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0731
Epoch 176/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0752
Epoch 177/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0762
Epoch 178/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0716
Epoch 179/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0717
Epoch 180/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0720
Epoch 181/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0784
Epoch 182/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0721
Epoch 183/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0642
```

```
Epoch 184/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0675
Epoch 185/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0659
Epoch 186/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0698
Epoch 187/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0800
Epoch 188/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0657
Epoch 189/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0644
Epoch 190/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0657
Epoch 191/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 0.0611
Epoch 192/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0678
Epoch 193/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0593
Epoch 194/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0689
Epoch 195/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0642
Epoch 196/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0648
Epoch 197/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 15ms/step - loss: 0.0641
Epoch 198/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0588
Epoch 199/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - loss: 0.0738
Epoch 200/200
3/3 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - loss: 0.0681

Generated Text (Transformer):
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 846ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step
deep learning improves sequence modeling performance used used
```

```
print("\nGenerated Text (Transformer):")
print(generate_text_transformer("long short term", 6))
```

```
Generated Text (Transformer):
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 100ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 63ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 79ms/step
long short term memory networks solve long dependency problems
```